Test Plan — SA400 Drive-Select / Write-Gate LED Monitor

Document: TEST_PLAN.md · Project: `sa400_led_monitor` · Test-plan version **1.2** · created 2026-10-19 · updated 2026-10-19

Revisions: 1.1 populated-board drawing and legend (§0.7) · 1.2 design-review findings folded in (§0.8, 2.6.17, 2.7.2, 2.7.4, 3.2.5, 3.3.7, §4, §5.3)

Colour key: **red** = a stop condition, a limit not to exceed, or a correction to an earlier document · **blue** = new or changed in revision 1.2 (the 2026-10-19 design review)

Applies to: board rev **v0.2.2** copper (the 0.2.3 gerber changed Q1 silkscreen only), populated by JLCPCB from the **r2** BOM/CPL in `jlcpcb_v0.2.2/`.

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0. Read this first

0.1 What the three phases prove

Phase	Setup	What it proves	Can it damage the TRS-80?
1 — Unpowered	Board only, DMM only	Assembly is right (orientation, values, no shorts); the bus fingers connect only where the design says	No — nothing is powered
2 — Bench power	Board + current-limited bench supply + a simple bus-simulator jig	Logic and LEDs work; the protection works; the **most** current and voltage the board can ever put on a bus line, measured	No — the EI and drives are not connected
3 — On the EI	Board on the real cable with the Expansion Interface, then with drives	Works with real signals; does not disturb disk I/O; safe in every power-on/power-off order	Only if a Phase 1 or 2 check was skipped or failed

0.2 Rules

1. **Do the phases in order, on each board.** A board does not go to the next phase until every check in the current phase passes.
2. **Stop on the first out-of-spec reading.** Power off, write it down, and fix or diagnose before going on. Do not “see if it works anyway” with the EI attached.
3. **Never plug or unplug the card edge (J1) with anything powered.** Power (J2) may be unplugged while running only in step 3.5, which is designed for it.
4. **Board 1 goes all the way through first.** Boards 2–5 do Phase 1 and Phase 2 sections 2.1–2.6. Only board 1 does the protection tests in 2.7 and 2.8 (they stress parts on purpose).
5. Write a serial number (1–5) on each board’s silkscreen with a marker, and record results in the sheets in section 5.

0.3 As-built values (r2 BOM): these differ from older docs

`docs/BOM.md` and `THEORY_OF_OPERATION.md` still show the design-time values. The boards were built from the r2 BOM, so **this plan uses the r2 values**:

Ref	Design docs say	**As built (r2)**	Effect on readings
R1–R8 (drive LEDs), R21 (5V LED)	1.5 kΩ	**470 Ω**	LEDs run ≈ 5.5–6.5 mA (still inside the 74LS00’s 8 mA sink rating)
R19 (12V LED)	4.7 kΩ	**2.2 kΩ**	D12 runs ≈ 4.5 mA

Ref	Design docs say	**As built (r2)**	Effect on readings
Q1	NDP6020P	**IRF4905PBF** (same G-D-S pinout)	Specified for V_GS = −10 V; at our −5 V it should still be fully on at 35 mA, and **step 2.3 measures that**
R9–R13, R14–R18	10 kΩ, 470 Ω	same	—

0.4 Reference map (used throughout)

J1 card edge. Even pins are on the component side, odd pins on the solder side directly underneath.

Signal	J1 pin	Its GND partner (under it)	Series R (finger → receiver)	Pull-up	U1 input	U1 output	Red LED / driver	Green LED / driver
DS1 (TRS-80 drive :0)	10	9	R17	R9	U1-1	U1-2	D1 / U3-2	D5 / U2-3
DS2 (drive :1)	12	11	R16	R10	U1-3	U1-4	D2 / U3-4	D6 / U2-6
DS3 (drive :2)	14	13	R15	R11	U1-5	U1-6	D3 / U3-6	D7 / U2-8
DS4 (drive :3)	6	5	R18	R12	U1-9	U1-8	D4 / U3-8	D8 / U2-11
WRITE GATE	24	23	R14	R13	U1-11	U1-10	—	(all greens through U2)

- **GND fingers (15):** 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31.
- **No-connect fingers (14):** 1 and 33 (they sit over the clip holes), and 2, 4, 8, 16, 18, 20, 22, 26, 28, 30, 32, 34.

J2 power (Mate-N-Lok, solder side, pin 1 = square pad). 1 = +12 V, 2 = +12 RTN, 3 = +5 RTN, 4 = +5 V. This is the same connector and pinout as a PC “Molex” 4-pin peripheral plug (yellow +12, black, black, red +5).

Power path. J2-4 → F1 (PTC) → Q1 pin 2 (drain) → Q1 pin 3 (source) = **+5V rail** (also U1/U2/U3 pin 14, D11 cathode, C4+). Q1 pin 1 (gate) = GND. J2-1 → R19 → D12 → GND only.

Handy test points. The vertical resistors stand up with one lead looped over the top. That loop is a good probe point, but which net it is on depends on which way JLCPCB stood the part up. **Before you use any resistor loop as a test point, check it with the DMM against the finger or the IC pin.**

0.5 Equipment

Item	Needed for	Notes
DMM with ohms, diode, DC volts, DC mA	all	A second DMM to watch current while you probe saves time
Bench supply, adjustable 0–15 V, **adjustable current limit**	2, 3	Two outputs is ideal (one for the board, one for the jig pull-ups). The outputs must be floating (not tied to earth on +)
Power pigtail for J2	2, 3	TE/AMP 1-480424-0 housing + 60619-1 pins — https://www.te.com/en/product-1-480424-0.html · pins https://www.mouser.com/c/?q=60619-1 — or any PC Molex-to-bare-wire cable (same connector). Label the wires.
Bus-simulator jig (section 0.6)	2	Recommended. The fallback is clip leads on the fingers, which risks scratching the gold or bridging two fingers
IC extractor, ESD strap	1, 2.7, 2.8	ICs are socketed, so you can pull them for the protection tests

Item	Needed for	Notes
Scope (optional)	3.3	Best way to show the board doesn't change bus levels or edges
IR thermometer or fingertip	2, 3.6	Heat check
Scratch diskettes (2+)	3	**Never test writes on a disk you care about**

0.6 Bus-simulator jig ("fake EI")

This stands in for the EI's open-collector drivers and the terminating drive. Each line is held high by 150 Ω (like the terminator), and a switch pulls it to ground (like a 7438 asserting the line).

```
3M 3463-0001 card-edge IDC  --  ~150 mm 34-way ribbon  --  cut end; separate:
  (red stripe = pin 1)            signals : 6, 10, 12, 14, 24
                                  GND     : 5, 9, 11, 13, 23 (twist together)

Each signal conductor (x5):

  +5V_JIG --[JP on/off]--[150 R]--+-- conductor 6 / 10 / 12 / 14 / 24
                                  |
                                [SW n]   closed = ASSERTED
                                  |
                                 GND --- GND conductors + bench supply (-)
```

- Use a 3463-0001 from the cable build: https://www.mouser.com/c/?q=3463-0001
- Ribbon conductor N = J1 pin N when the stripe is on pin 1 of the IDC connector. Before you trust the jig, check that with the DMM.
- **JP** turns the five 150 Ω pull-ups on or off. *Off* = the board's own 10 k pull-ups only. *On* = simulates a terminated bus. +5V_JIG comes from the second bench output (or the same output, but then the board-current reading also includes the jig current).
- Five switches, labelled DS1 DS2 DS3 DS4 WG.

0.7 Populated board (wireframe) and legend

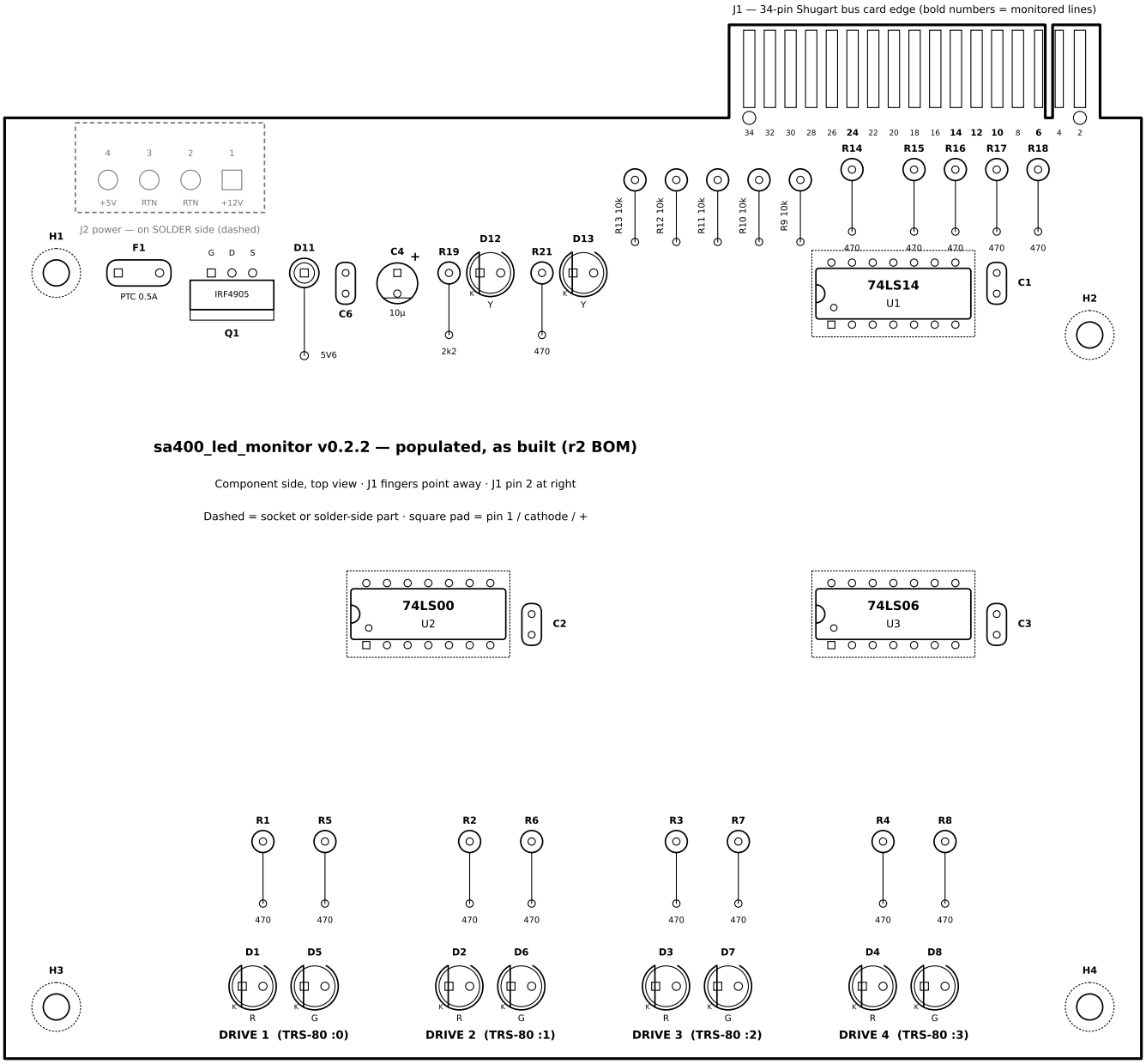


Figure 1: Populated board, component side, wireframe — generated by gen/render_populated.py from the footprint positions in gen/design.py, as-built values

Ref	Part (as built, r2 BOM)	Drawn as	Job on the board	Check when inspecting
U1	74LS14 hex Schmitt inverter, DIP-14, in socket	Solid rounded box with end notch, inside a dashed socket outline	Bus receivers for DS1–DS4 and WRITE GATE	Notch at left; pin 1 = square pad, bottom-left
U2	74LS00 quad NAND, DIP-14, in socket	Same	Green LEDs: DSn AND WG	Same
U3	74LS06 hex open-collector inverter, DIP-14, in socket	Same	Red LEDs: DSn	Same
Q1	IRF4905 P-channel MOSFET, TO-220, standing	Rectangle (body) + thin rectangle (metal tab), pins G D S	Reverse-polarity switch on +5 V	Gate in the square pad; tab toward the front edge
F1	PTC resettable fuse 0.5 A	Rounded bar	+5 V input fuse	—
D11	1N5339B 5.6 V 5 W Zener, DO-15, standing	Double circle (body + band) with the lead down to the anode	+5 V rail clamp	Band (cathode) on top, in the square pad
C4	10 µF 16 V electrolytic	Circle with “+” and a line on the stripe side	Bulk +5 V	+ lead in the square pad
C1, C2, C3, C6	100 nF ceramic disc	Small rounded slot	Decoupling U1/U2/U3; rail entry (C6)	—
R14–R18	470 Ω ¼ W, standing	Circle (body) + line (lead)	Bus series resistors, finger → receiver (fault-current limit)	Value printed under the lead
R9–R13	10 kΩ ¼ W, standing	Same, label rotated	Bus pull-ups on the receiver side	—
R1–R8	470 Ω ¼ W, standing	Same	Drive-LED series resistors (R1–R4 red, R5–R8 green)	—
R19	2.2 kΩ ¼ W	Same	+12 V indicator series	—
R21	470 Ω ¼ W	Same	+5 V indicator series	—
D1–D4	Red 5 mm LED (R)	Circle with a flat; “K” at the flat	Drive 1–4 SELECTED	Flat / short lead = cathode = square pad
D5–D8	Green 5 mm LED (G)	Same	Drive 1–4 WRITE GATE	Same
D12, D13	Yellow 5 mm LED (Y)	Same	“12V” and “5V” present	Same
J1	34-contact card edge (gold fingers)	Rectangles on the tab; bold numbers = monitored lines 6/10/12/14/24	Shugart bus	Even pins on top, odd under them; key slot between 4 and 6
J2	TE/AMP Mate-N-Lok 350211-1	**Dashed grey**: it sits on the solder side	Power in: 1 +12 V, 2/3 RTN, 4 +5 V	Pin 1 = square pad
H1–H4	M3 mounting holes	Circle in a dashed ring	Standoffs	—

Conventions: a **square pad** is pin 1, a cathode, or a capacitor’s + lead. **Dashed** means a socket or a part on the solder side. Drive *n* on the board is TRS-80 drive *:n−1*.

0.8 Design-review findings (2026-10-19) and where each one is tested

Added in 1.2. A review of the v0.2.2 design against the datasheets found one real margin issue (F1) and three smaller points. None of them stops the test run; each is tied to a step below.

#	Finding	Tested at	What to do
F1	**Receiver low-level margin.** When a bus line is low, the LS14 input current (0.4 mA max) *and* the R9–R13 pull-up current (≈ 0.44 mA) both flow out through the 470 Ω series resistor. Worst case the U1 input sits at ≈ 0.79 V (0.4 V VOL + 0.84 mA × 470 Ω) against a 74LS14 VT− of 0.5 V min / 0.8 V typ / 1.0 V max. Typical parts give ≈ 0.55 V, so it works on typical margin. THEORY §3.0's 0.59 V figure left out the pull-up term; corrected there.	**3.3.7** (the number that matters), 3.2.5, 2.6.17	If 3.3.7 reads > 0.5 V: fit a **74HCT14** in U1's socket (no soldering; input current is µA, which removes ≈ 0.19 V). Optionally R9–R13 → 47–100 k (removes most of the other ≈ 0.2 V). v0.3: pull-ups move to the finger side of R14–R18 (V0.3_PLAN).
F2	**+12 V on the +5 V pin** is harsher than THEORY §4.3 said. A cold 0.5 A PTC is ≈ 0.5 Ω, so until F1 trips D11 takes whatever the supply will deliver at ≈ 6 V. F1 trips and the EI never sees it, but D11 may fail (normally short: board dead, still safe).	2.7.4	Stays optional. Expect to lose D11 if you run it from a stiff supply. The Mate-N-Lok is keyed; this fault needs a hand-made cable.
F3	**D12 reverse rating.** Reversed +12 V puts ≈ 7 V reverse across D12 (5 mm LEDs are rated ≈ 5 V) at ≈ 3 mA through R19. Likely survives, but out of spec.	2.7.2	Run it knowing that, or skip it. v0.3: 1N4148 in series with D12.
F4	Minor: D11 (5.6 V, soft knee) leaks a few mA at 5.25 V and runs slightly warm; U2 sinks ≈ 5.3 mA per green LED with the as-built 470 Ω (LS00 rating 8 mA); IRF4905 at V_GS = −5 V.	2.4.2, 2.6.20, 2.3	Record. The 74LS37 is a pin-compatible 24 mA drop-in for U2 if brighter greens are wanted later.

Checked in the same review and found right: Q1's orientation (drain in, source out, gate to ground is the correct way round for a P-channel reverse-polarity switch), all unused inputs tied, no path from +12 V to either the rail or the bus, ≤ 12 mA worst case into an EI line, no termination, the LS06 for the reds, and U1's fan-out.

Phase 1 — Unpowered (no supply, not on the EI)

Board fully assembled, ICs in their sockets, nothing plugged into J1 or J2. Hold the DMM leads still while a reading settles, because the capacitors charge through the meter.

1.1 Visual and mechanical

#	Check	Pass
1.1.1	U1 = 74LS14, U2 = 74LS00, U3 = 74LS06, each in the correct position, notch matches the silk, every pin inside the socket (none folded under)	✓
1.1.2	Q1 is the IRF4905: pin 1 (gate) in the square pad, metal tab on the side shown by the silkscreen	✓
1.1.3	D11: band (cathode) at the square pad. C4: + lead at the square pad	✓
1.1.4	All LEDs: flat side / short lead (cathode) at the square pad. Colours: D1–D4 red, D5–D8 green, D12–D13 yellow	✓
1.1.5	Resistor colour bands: R1–R8, R21, R14–R18 = 470 Ω (yellow-violet-brown); R9–R13 = 10 k (brown-black-orange); R19 = 2.2 k (red-red-black-brown for 1 %)	✓
1.1.6	J2 is on the **solder side**, housing opening toward the rear edge, seated flat	✓
1.1.7	Solder side: no bridges, especially across the J2 pins and the DIP rows. No flux balls	✓
1.1.8	Fingers: gold, bevelled, **no solder or flux on any finger**, no lifted finger	✓
1.1.9	Board thickness at the fingers (calipers): 1.37–1.78 mm (3M spec), expect 1.6	record
1.1.10	Fit test: push a **bare** 3463 connector (not on a live cable) onto the tab. It seats fully, and the key (if fitted) goes into the slot between pins 4 and 6. Remove it	✓
1.1.11	Fit test: the power plug latches into J2 and only goes in one way	✓
1.1.12	**Record the three ASSUMED dimensions** while you have calipers out: tab edge to corner (5.08 assumed), key-plug width (0.91), J2 pin drill (2.1) — see `docs/MECHANICAL.md`	record

1.2 Power connector — no shorts between rails

DMM in ohms unless stated. “OL” = open / over-range.

#	From (red)	To (black)	Expect	Proves
1.2.1	J2-1 (+12)	J2-4 (+5)	**OL**, then try the leads the other way: **OL**	12 V can never reach the 5 V rail through the board
1.2.2	J2-2	J2-3	< 1 Ω	Both returns are tied together
1.2.3	J2-3	J1-9 (any GND finger)	< 1 Ω	Supply ground = bus ground
1.2.4	J2-4	J2-3 (GND)	> 200 Ω and settling (not a dead short)	No short across the +5 V input
1.2.5	J2-1	J2-3	> 2 kΩ or OL	No short across the +12 V input
1.2.6	J2-4	F1 far lead / Q1 pin 2	< 2 Ω (cold PTC)	F1 is in circuit and intact

#	From (red)	To (black)	Expect	Proves
1.2.7	Q1 pin 1 (gate)	GND	< 1 Ω	Gate tied to ground as designed

1.3 Reverse-polarity switch (Q1) orientation — diode mode

#	Red lead	Black lead	Expect	Proves
1.3.1	Q1 pin 2 (drain)	Q1 pin 3 (source / rail)	0.4–0.7 V	Body diode points from input to rail, so Q1 is fitted the right way round
1.3.2	Q1 pin 3	Q1 pin 2	OL (may start low and climb while C4 charges)	Reversed supply will be blocked
1.3.3	J2-4	U1-14 (+5V rail)	0.4–0.7 V	Whole input path F1 → Q1 → rail is in one piece
1.3.4	GND	U1-14	0.4–0.8 V	Normal: D11 and the IC substrate diodes. **≈ 0 V = short on the rail**

If 1.3.1 reads OL and 1.3.2 reads 0.5 V, Q1 is in backwards. **Stop.**

1.4 Bus fingers — the board touches only what it should

This is the core “cannot hurt the EI” check with no power applied.

#	Test	Expect
1.4.1	Each of the **14 no-connect fingers** (1, 33, 2, 4, 8, 16, 18, 20, 22, 26, 28, 30, 32, 34) to GND (J2-3)	**OL**
1.4.2	The same 14 fingers to the +5V rail (U1-14)	**OL**
1.4.3	Each of the **15 GND fingers** to J2-3	< 1 Ω
1.4.4	Each monitored finger to its U1 input (table 0.4, e.g. J1-10 → U1-1)	**447–494 Ω** (470 Ω ±5 %)
1.4.5	Each monitored finger to the +5V rail, red lead on the finger	**9.9–11.0 kΩ** (470 Ω + 10 kΩ)
1.4.6	Each monitored finger to GND	> 10 kΩ, **never < 1 kΩ**
1.4.7	Adjacent signal fingers to each other: 10↔12, 12↔14, 4↔6, 6↔8, 22↔24, 24↔26	≈ 21 kΩ between two monitored fingers (each one’s 10.47 k to the rail), **OL** when one of the pair is a no-connect
1.4.8	Each monitored finger and J2-1 (+12)	**OL** — 12 V has no path to the bus

1.5 LEDs in circuit — diode mode

Red lead on the anode (round pad), black lead on the cathode (square pad). Most DMMs light a red or yellow LED faintly. Some can’t light green.

#	LED	Expect
1.5.1	D1–D4, D12, D13	1.5–2.1 V, faint glow
1.5.2	D5–D8	1.7–2.4 V or OL (meter can’t reach a green’s Vf); faint glow if it can
1.5.3	Each LED reversed	OL or a much higher reading (other in-circuit paths)

Phase 1 done → record, go to Phase 2.

Phase 2 — Bench power, not on the EI

Board on the bench, J2 fed from the bench supply through the pigtail. J1 is either empty or plugged into the **jig only**. The EI, drives and cable are nowhere near it.

Expected currents (as built):

Condition	Expected +5 V current
Idle (no line asserted)	**≈ 25–45 mA** (three LS ICs + D13 at ≈ 6 mA) — accept 15–60 mA, but boards should agree within ±20 %
Each asserted line (jig pull-ups off)	+ ≈ 0.48 mA (its 10.47 k pull-up path) + any LED it lights
One drive selected	+ ≈ 5.5–6 mA (red)
Selected and writing	+ another ≈ 5.5 mA (green)
+12 V input	≈ 4.5 mA (D12 only)

2.1 First power-up — +5 V only, current-limited

1. Supply output **off**. Set it to **0 V, 100 mA limit**. Connect J2-4 (+) and J2-3 (−). Leave J2-1 open. Nothing on J1.
2. Turn the output on and **raise slowly** to 5.00 V, watching the current.
   - Over 60 mA at any point, or the supply drops into current limit → **stop**: short, or a reversed IC.
3. At 5.00 V record:

#	Measure	Expect
2.1.1	Supply current	15–60 mA (record exactly — this is the board's fingerprint)
2.1.2	D13 "5V"	lit
2.1.3	D1–D8, D12	**all off** (no line asserted; D12 has no +12 V)
2.1.4	Touch test after 1 min: U1–U3, Q1, D11, F1	none warm

2.2 Rail voltages

#	Measure (black on GND)	Expect
2.2.1	J2-4	5.00 V (set)
2.2.2	+5V rail at U1-14, U2-14, U3-14	within 0.05 V of J2-4
2.2.3	U1-7, U2-7, U3-7	0.00 V

2.3 Q1 voltage drop

#	Measure	Expect	If not
2.3.1	DC volts, red J2-4, black U1-14	**< 0.05 V**	0.4–0.7 V means Q1's channel is not turning on and the current is going through its body diode. The rail would then sit around 4.4 V, below the 74LS minimum of 4.75 V. Check Q1 orientation and part; the IRF4905 should be fully on at −5 V V_GS for 35 mA

2.4 Supply tolerance and clamp leakage

#	Set J2-4 to	Measure	Expect
2.4.1	4.75 V	current; then repeat 2.5 for one drive	LEDs still work
2.4.2	5.25 V	current	No more than **5 mA** above the 5.00 V reading. A bigger rise means D11 (5.32 V minimum knee) is starting to conduct; record it
2.4.3	back to 5.00 V	—	—

2.5 +12 V indicator

1. Second output (or the same supply in a second step): 12.0 V, **50 mA limit**, J2-1 (+) / J2-2 (−).

#	Measure	Expect
2.5.1	+12 V current	3.5–5.5 mA
2.5.2	D12 “12V”	lit
2.5.3	+5V rail (U1-14)	unchanged from 2.2 — 12 V does not leak onto the logic
2.5.4	Every monitored finger (6, 10, 12, 14, 24), open circuit	≈ +5 V, **never above the rail**

2.6 Bus isolation and logic — with the jig

The monitor stays at 5.00 V / 100 mA limit. Plug the **jig** (pull-ups **off**, all switches open) onto J1 with the board’s supply **off**, then power up.

2.6.A — What the board puts on the bus (the “will it short the EI or a drive” numbers)

#	Test	Expect	Proves
2.6.1	DC volts, each monitored line to GND, all switches open	4.8–5.25 V (the rail through 10.47 k)	The board only ever *pulls up weakly*; it never drives a line
2.6.2	DC **mA** meter from each monitored line to GND (meter in place of the switch)	**0.45–0.55 mA**	The **most** current the board can push into a bus line in normal operation: < 2 % of the 31 mA the terminator already makes the EI sink
2.6.3	DC volts, each **no-connect** finger to GND (probe the ribbon conductor)	0.00 V (floating)	The board leaves the other 12 signals alone
2.6.4	Board supply current with all five switches closed (lines held low), no LEDs counted	idle + 5 × 0.48 ≈ idle + 2.4 mA (plus the LEDs that light)	Matches the pull-up model

2.6.B — Logic truth table (jig pull-ups off, then repeat with them **on**)

#	Switches closed	Expect lit	Must stay dark
2.6.5	none	D13	D1–D8
2.6.6	DS1	D1	all other drive LEDs
2.6.7	DS2	D2	”
2.6.8	DS3	D3	”
2.6.9	DS4	D4	”
2.6.10	WG only	none	D1–D8 (write gate with no drive selected lights nothing)
2.6.11	DS1 + WG	D1 + D5	others
2.6.12	DS2 + WG	D2 + D6	others
2.6.13	DS3 + WG	D3 + D7	others

#	Switches closed	Expect lit	Must stay dark
2.6.14	DS4 + WG	D4 + D8	others
2.6.15	DS1 + DS3 (two selects, not legal on a real bus)	D1 + D3	— shows the channels are independent

2.6.C — **Levels** (one channel, DS1 + WG closed, jig pull-ups **on**)

#	Measure	Expect
2.6.16	J1-10 (finger)	≈ 0 V (switch)
2.6.17	U1-1 (receiver input, after R17)	**< 0.45 V** (1.2: was 0.4 V. A hard ground on the jig gives ≈ 0.31 V typical, 0.40 V worst case: F1)
2.6.18	U1-2 (DS1, active-high)	> 2.7 V
2.6.19	U3-2 (red cathode) / U2-3 (green cathode)	< 0.5 V each
2.6.20	Voltage across R1 (red D1 resistor)	2.5–2.9 V → ≈ 5.3–6.2 mA
2.6.21	Switch open again: U1-1	> 4.5 V; U1-2 < 0.5 V; LEDs off

If a channel fails, follow it down that row of table 0.4: finger → series R → U1 input → U1 output → U3/U2 output → LED. The first stage that reads wrong is the fault.

2.6.D — **Unpowered board on a powered bus** (EI on, monitor off — the reverse case)

#	Test	Expect
2.6.22	Board supply **disconnected** from J2. Jig pull-ups **on** (+5V_JIG = 5.0 V). mA meter in series with one 150 Ω pull-up → line	Line current changes by **< 0.6 mA** between the board plugged in and the board off the jig
2.6.23	With all five pull-ups on, measure the dead board's +5V rail (U1-14)	Record. Expect well under 2 V; D13 may glow very faintly. The back-feed is limited by 470 Ω + 10 kΩ per line
2.6.24	Line high voltage at the jig with the dead board plugged in	≥ 4.9 V (the board doesn't drag the bus down when it's unpowered)

2.7 Protection tests — board 1 only

These deliberately apply the faults in `THEORY_OF_OPERATION.md` §4.3. **Pull U1, U2 and U3 from their sockets first** (note the orientation) so an unexpected result can only cost a cheap part. Jig disconnected.

#	Fault applied	How	Expect	Pass
2.7.1	**Reversed +5 V**	5.0 V, 50 mA limit, **+ on J2-3, − on J2-4**	Current < 1 mA; rail (U1 socket pin 14 to GND) ≈ 0 V; D13 dark	Q1 blocks
2.7.2	**Reversed +12 V**	12 V, 50 mA limit, + on J2-2, − on J2-1	Current ≈ 3 mA (D12 in reverse breakdown through R19); D12 dark	Nothing else conducts. Puts D12 ≈ 2 V past its reverse rating (F3). Likely survives; skip if you'd rather not.
2.7.3	**Overvoltage on +5 V** (clamp)	Normal polarity, **300 mA limit**. Raise J2-4 from 5.0 V in 0.2 V steps to 7.0 V or until the supply hits its limit. At each step record the input V, the current and the rail V	Rail follows the input up to D11's knee (≈ 5.3–5.9 V); after that the current climbs steeply while the rail rises slowly	Rail **< 6.5 V** at 300 mA (74LS absolute max is 7.0 V). Don't hold more than 30 s at 300 mA

#	Fault applied	How	Expect	Pass
2.7.4	*Optional* — **12 V on the +5 V pin** (miswired cable)	Normal polarity, **12 V, 2 A limit**, ICs still out. Watch the current; switch off after 10 s regardless	Current goes straight to the supply's 2 A limit (the cold PTC is ≈ 0.5 Ω; it does not limit anything), D11 dissipates ≈ 12 W until F1 trips, well under a second; then the current falls to tens of mA. Rail stays < 7 V throughout. Let it cool 5 min, then repeat 2.1 → normal	F1 trips; nothing smokes. D11 may fail short (F2): then the rail reads ≈ 0 V and D11 must be replaced before 2.1 passes.

Skip 2.7.4 unless you are willing to lose D11. From a 2 A bench limit it will probably survive the pulse; from a PC supply (tens of amps) it would not. Steps 2.7.1–2.7.3 already show the clamp and the reverse block work.

2.8 Worst-case fault current into a bus line — board 1 only

This checks the worst case the 470 Ω series resistors are there for: a dead receiver shorting a bus line to the rail.

1. ICs **out**. Board at 5.00 V, 100 mA limit.
2. Jumper **U1 socket pin 1 (DS1_in) to U1 socket pin 14 (+5 V)**. This simulates a receiver input welded to the rail.
3. mA meter from **J1-10** (finger) to **GND**.

#	Expect	Pass
2.8.1	≈ 10.6 mA (5.0 V / 470 Ω)	**≤ 12 mA** — added to the terminator's 31 mA, that's 43 mA, inside the 7438's 48 mA rating

Repeat for one more channel (e.g. U1 pin 11 / J1-24). Remove the jumper, refit the ICs (check the notches), and re-run 2.1 and 2.6.B quickly.

Phase 2 done → record, go to Phase 3 (board 1 first).

Phase 3 — On the Expansion Interface

3.0 Before starting

- ☐ Board has passed Phase 1 and Phase 2.
- ☐ **Baseline without the monitor:** Model I + EI + drives boot DOS, `DIR` each drive, read a few files. Write down anything that already misbehaves, so it isn't blamed on the monitor later.
- ☐ **Cable.** Use the six-connector cable from `docs/CABLE.md`, or a straight-through cable with a spare connector. **On a Radio Shack "pulled-pin" cable** each drive connector carries only one drive-select line, so the monitor would only ever see that one drive. Put it on a connector with all 34 contacts.
- ☐ Cable continuity: each connector, pin N to pin N, and no adjacent-pin shorts (DMM, 2 minutes).
- ☐ Red stripe = pin 1 on every connector. On the monitor, pin 1 is the solder-side finger directly under pin 2, at the **board-corner end of the tab, away from J2**. Confirm against the silk pin numbers before mating.
- ☐ The terminated drive is at the **far end** of the cable. The monitor adds no terminator.

3.1 Ground and supply check

1. Model I + EI **on**, cable connected to the EI but **not** to the monitor. Monitor supply on but not connected to the monitor.
2. DC volts between the **bench supply (−)** and the **EI chassis / a GND pin**: < 0.1 V. AC volts: < 0.1 V.
   - A higher reading means the bench supply or the EI is floating at a different potential. Connecting it through the cable's ground would push current through the ribbon. **Fix first** (bring both to the same mains ground, or use a floating supply).
3. Everything **off**.

3.2 Stage A — EI + monitor, no drives

1. Everything off. Plug the monitor onto the DIAG connector (stripe to pin 1). **No drives connected.**
2. Monitor supply: 5.00 V, 100 mA limit (+12 V optional). Power on → idle current same as 2.1.1 (± 2 mA).
3. Power on the Model I / EI.

#	Observe	Expect
3.2.1	Monitor current as the EI powers up	stays within idle + 15 mA, no spike to the limit
3.2.2	At power-on, with no drive fitted	The ROM tries to boot drive :0, so **D1 should light** (DS1 asserted). Hold BREAK (or reset with BREAK) to get to BASIC
3.2.3	Run the select-cycling program below	D1 → D2 → D3 → D4 in turn, about 1 s each; **no green ever**
3.2.4	DMM on J1-10 side of R17 (or scope) while :0 selected	≤ 0.4 V asserted, ≈ 5 V released
3.2.5 (1.2, F1)	DMM on **U1-1** (the DS1 receiver input, after R17) while :0 is selected: hold BREAK at power-on, or the cycling program	Record. With no drives there is no terminator, so the EI driver sinks almost nothing and this reads low, ≈ 0.3–0.4 V. The number that matters is 3.3.7.

```
10 REM CYCLE DRIVE SELECTS :0-:3, 37E1H = 14305 (MODEL I DRIVE SELECT LATCH)
20 FOR D = 0 TO 3
30 V = 2 ^ D
40 FOR T = 1 TO 150 : POKE 14305, V : NEXT T
50 NEXT D
60 GOTO 20
```

The EI's select latch is cleared by the motor-on timeout after a few seconds, which is why line 40 keeps re-writing it. Without drives there's no terminator, so the monitor's 10 k pull-ups are the only thing pulling the lines high. That's fine for this test.

3.3 Stage B — EI + monitor + drives

1. Everything off. Connect the drives (terminated drive at the end). Monitor on DIAG.
2. Power order: monitor supply → drives → Model I / EI.

#	Test	Expect
3.3.1	Boot DOS from :0	Boots exactly as in the baseline. D1 lights whenever drive :0's own activity LED does
3.3.2	`DIR` on each drive present	Red LED for that drive only, in step with the drive's own LED
3.3.3	Run the BASIC select-cycling program (from Disk BASIC)	As 3.2.3; the drive motors start, which is normal
3.3.4	**Write test** — scratch disk in :1. Save a file or format it (syntax depends on the DOS)	D2 red throughout, **D6 green only while writing**. No green on any other drive

#	Test	Expect
3.3.5	Read back / `DIR` the scratch disk; copy a file :0 → :1 and compare	No errors, same as the baseline
3.3.6	*(scope, optional)* DS1 at the DIAG connector, monitor plugged in vs removed (everything off to swap)	Low level differs by < 50 mV; rise and fall times look the same
3.3.7 (1.2, F1)	**Receiver margin.** With the drives connected (terminated bus), DMM on **U1-1** while :0 is selected (repeated `DIR :0`, or the cycling program). Also U1-3 / U1-5 / U1-9 for :1 / :2 / :3 if those drives are fitted. Note the finger voltage (J1-10) alongside for comparison.	**≤ 0.50 V = clean** (below the 74LS14's minimum VT−). **0.50–0.80 V = works on typical margin**: fit a 74HCT14 in U1 (and optionally 47–100 k for R9–R13), re-measure; each change should drop the reading by ≈ 0.2 V. **> 0.80 V = a select can be missed.** Do not rely on the board until U1 is swapped and the reading is ≤ 0.8 V.

3.4 Power-order robustness

Run a read loop (e.g. repeated `DIR :0`, or a BASIC program that LOADs a file over and over) and, **without touching the card edge**:

#	Action	Expect
3.4.1	Unplug the monitor's **J2** (power) while the loop runs	Loop continues with no errors; monitor LEDs go dark
3.4.2	Plug J2 back in	Loop continues; LEDs resume
3.4.3	Monitor powered, Model I / EI and drives off	Monitor current back to idle; no LEDs except D13 (and D12)
3.4.4	Monitor unpowered, power the system on and boot	Boots normally (an unpowered monitor doesn't load the bus)

3.5 Soak

#	Test	Expect
3.5.1	1 hour of mixed read/write on a scratch disk	No disk errors beyond the baseline
3.5.2	Heat check at the end: U1–U3, Q1, D11, F1, R1–R8	All barely warm (< 45 °C). **D11 warm** means the rail is above its knee: check the supply voltage
3.5.3	Monitor current at the end	Same as at the start (± 2 mA)

3.6 Final power source

If the monitor will normally be powered from something other than the bench supply (a PC PSU Molex lead, a drive's supply), repeat **2.1–2.2** on that source with a meter in series (expect 4.75–5.25 V at J2-4), then **3.3.1–3.3.4** briefly.

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4. If something fails

Symptom	Most likely	Look at
Current at the limit on first power-up	Reversed IC, solder bridge, C4 reversed	1.1.1, 1.1.3, 1.3.4
Rail ≈ 0.5 V below J2-4	Q1 conducting only through its body diode	2.3, Q1 orientation / part
All LEDs dead, D13 lit	U1 missing/reversed, or the fingers not making contact	U1-14 voltage, 2.6.C
One red never lights	That row of table 0.4	Series R, pull-up, U1 gate, U3 gate, LED polarity

Symptom	Most likely	Look at
Red lights but its green never does	That U2 gate, its LED, or WG path (R14, U1-11/10)	2.6.C
Green lights with its red even when not writing	WG stuck active: U1-11 shorted to GND, or U1 gate 5 faulty	U1-11 should be > 4.5 V with WG released; U1-10 < 0.5 V
Green never lights on any drive	WG path open: R14, or U1-11/U1-10	2.6.C on the WG row
LEDs flicker with nothing asserted	Pull-up open, so a floating LS14 input	R9–R13
A drive's red is erratic or misses short accesses that the drive's own LED shows	Receiver margin (F1): U1 input node too high	3.3.7; 74HCT14 in U1
Disk errors with the monitor fitted but not without	Cable (the sixth connector), terminator position, or a monitored line loaded	3.0 cable checks, 2.6.1–2.6.2 on that board, 3.3.6

5. Record sheets

Tick ✓ or write the reading. One column per board.

5.1 Phase 1

Step	B1	B2	B3	B4	B5
1.1.1–1.1.8 visual					
1.1.9 thickness (mm)					
1.1.10–1.1.11 fit					
1.1.12 tab-corner / key / J2 drill (mm)					
1.2.1 +12↔+5 OL					
1.2.4 J2-4↔GND (Ω)					
1.2.6 F1 (Ω)					
1.3.1 Q1 body diode (V)					
1.3.4 GND↔rail diode (V)					
1.4.1–1.4.2 no-connect fingers OL					
1.4.3 GND fingers					
1.4.4 series R (Ω) 10/12/14/6/24					
1.4.5 finger→rail (kΩ) 10/12/14/6/24					
1.4.8 fingers↔+12 OL					
1.5 LEDs					

5.2 Phase 2

Step	B1	B2	B3	B4	B5
2.1.1 idle current (mA)					
2.2.2 rail (V)					
2.3.1 Q1 drop (V)					
2.4.1 works at 4.75 V					
2.4.2 current at 5.25 V (mA)					
2.5.1 +12 current (mA)					
2.6.1 line open V (min/max)					
2.6.2 line short mA (max)					
2.6.5–2.6.15 truth table, pull-ups off					
2.6.5–2.6.15 truth table, pull-ups on					
2.6.20 LED resistor V					

Step	B1	B2	B3	B4	B5
2.6.22 dead-board line current (mA)					
2.6.23 back-fed rail (V)					
2.7.1 reverse +5 (mA)		—	—	—	—
2.7.3 rail at 300 mA (V)		—	—	—	—
2.7.4 PTC trip (optional)		—	—	—	—
2.8.1 fault current (mA)		—	—	—	—

5.3 Phase 3

Step	B1	B2	B3	B4	B5
3.1 ground offset (V DC / AC)					
3.2.1–3.2.4 EI only					
3.2.5 U1-1, no drives (V)					
3.3.1–3.3.5 with drives					
3.3.7 U1-1, with drives (V) → HCT14?					
3.3.6 scope delta (mV)					
3.4.1–3.4.4 power order					
3.5 soak / heat / current					
3.6 final supply					

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6. TODO

- ☐ Build the bus-simulator jig (0.6)
- ☐ Board 1: Phase 1 → Phase 2 (all, including 2.7/2.8) → Phase 3
- ☐ Boards 2–5: Phase 1 → Phase 2 (2.1–2.6) → Phase 3 (3.3.1–3.3.4)
- ☐ 3.3.7 result → decide whether U1 becomes a 74HCT14 (F1); record which boards got it
- ☐ Feed the 1.1.12 measurements into `gen/config.py` and close the ASSUMED items in `docs/MECHANICAL.md`
- ☐ Bring `docs/BOM.md` and `THEORY_OF_OPERATION.md` up to the r2 values (table 0.3)
- ☐ Results go into the v0.3 plan (`docs/V0.3_PLAN.md`)